



Research Paper: *AI Data Management & Quality Assurance*

Topic: *The Role of AI in Data Lifecycle Management*

Course: *AI Data Management and Quality Assurance (ARTI 409)*

Instructor: *Tim Cruz*

Group Members: *Roy Aggarwal, Neil McGuffin, Chuck Burns, Gurpreet Bajwa*

Submission Date: *June 4, 2026*

1. Introduction	3
2. Methodology.....	3
3. Findings: Benefits and Drawbacks of AI in the Data Management Lifecycle	4
3.1 Benefits	4
Collection, Preparation, and Classification	5
Storage, Monitoring, and Governance	6
Retention, Archiving, and Disposal	6
3.2 Drawbacks	7
Collection, Preparation, and Classification	7
Storage, Monitoring, and Governance	9
Retention, Archiving, and Disposal	10
4. Findings: Governance of AI in the Data Management Lifecycle	12
Collection, Preparation, and Classification	12
Storage, Monitoring, and Governance	13
Retention, Archiving, and Disposal	14
4.1 Recommendations	14
5. Conclusion	15
6. References	17
7. AI Usage Statement	17

1. Introduction

As organizations handle increasingly massive volumes of data, managing that information manually becomes less practical. This research paper examines "The Role of AI in Data Lifecycle Management" to understand how automated tools impact modern data engineering [1].

To provide a balanced view, the paper breaks the topic down into three core areas: the operational benefits, the technical drawbacks, and the necessary governance strategies across a specific nine-stage data lifecycle. These areas were examined through a qualitative literature review that draws on peer-reviewed research, international standards, government frameworks, and vendor documentation, as described in the Methodology section. By analyzing these areas, the report outlines how organizations can leverage AI efficiency while maintaining strict human-centric oversight and compliance [1], [2], [3].

2. Methodology

This research employed a qualitative literature review methodology to examine the role of Artificial Intelligence (AI) in Data Lifecycle Management. The study focused on identifying how AI technologies are applied across various stages of the data lifecycle, evaluating both operational benefits and associated risks, and assessing governance mechanisms required to ensure responsible implementation.

Information was gathered from a combination of peer-reviewed academic literature, international standards, government frameworks, and industry documentation. Key sources included the NIST Artificial Intelligence Risk Management Framework (AI RMF 1.0), ISO/IEC 25012 Data Quality Model, academic research on data cascades and explainable AI, and technical documentation from Microsoft, Google Cloud, and Amazon Web Services. These sources were selected due to their relevance to AI governance, data quality assurance, lifecycle management, and enterprise-scale implementation practices.

The analysis was structured around a nine-stage data lifecycle consisting of Data Collection, Data Preparation, Data Classification, Data Storage, Data Quality Monitoring, Data Governance, Data Retention, Data Archiving, and Data Disposal. For each lifecycle stage, the research evaluated three dimensions: (1) operational benefits provided by AI technologies, (2) technical and organizational challenges associated with AI adoption, and (3) governance controls required to mitigate identified risks. Findings were synthesized to provide a balanced assessment of AI's role as both an enabler of efficiency and a source of emerging governance concerns.

This methodology supports a comprehensive examination of AI-enabled lifecycle management by combining theoretical research, practical industry applications, and internationally recognized governance standards.

3. Findings: Benefits and Drawbacks of AI in the Data Management Lifecycle

3.1 Benefits

The benefit of using AI across the data lifecycle is that it turns data management from a mostly manual, periodic activity into a more continuous, metadata-driven process [4]. In practice, this means faster discovery, richer context, more consistent classification, and earlier anomaly detection.

There are multiple interpretations of the data lifecycle within industry and academic literature. This research follows a nine-stage data lifecycle consisting of Data Collection, Data Preparation, Data Classification, Data Storage, Data Quality Monitoring, Data Governance, Data Retention, Data Archiving, and Data Disposal. Each stage is examined to assess the operational benefits, potential risks, and governance considerations associated with AI-enabled data management.

The discussion below also references industry tools provided by the large hyper-scale public cloud providers Microsoft, Google, and Amazon that can be used at various stages. The AI-enabled data governance and lifecycle management tools discussed are Amazon Macie for sensitive-data discovery in Amazon Simple Storage Service (S3) [4], Microsoft Purview for ML-assisted classification and lifecycle management in Microsoft 365 [5], and Google Cloud Knowledge Catalog for metadata harvesting, lineage, profiling, and quality checks for data distributed across various locations within organizations [6].

Table 1: AI Applications Across the Data Lifecycle

Data Lifecycle Stage	Primary AI Benefit	Example Technology
Data Collection	Automated metadata harvesting and sensitive-data detection	Google Knowledge Catalog, AWS Macie
Data Preparation	Automated profiling, cleansing, and enrichment	Google Cloud Data Quality
Data Classification	Intelligent tagging and classification	Microsoft Purview, AWS Macie
Data Storage	Improved discoverability and indexing	AWS Macie
Data Quality Monitoring	Continuous anomaly detection and validation	Google Cloud Data Quality
Data Governance	Policy enforcement and lineage management	Microsoft Purview
Data Retention	Lifecycle policy management	Microsoft Purview
Data Archiving	Metadata preservation and retrieval optimization	AI-enabled archival systems
Data Disposal	Identification of deletion candidates and audit support	Microsoft Purview

Collection, Preparation, and Classification

Data Collection: AI can provide clear benefits at the collection stage by intelligently ingesting data, validating data, capturing/extracting metadata, and flagging risk much earlier than manual intake processes. Google Cloud’s Knowledge Catalog can automatically harvest technical metadata across foundational systems and unify it into a governed source of truth [6], while AWS’s Macie product can be inserted into data-ingestion pipelines to detect sensitive data before it advances deeper into a lake or analytics workflow [4]. This reduces blind spots at intake, speeds onboarding of new sources, and lowers the chance that poorly understood or sensitive records enter downstream systems unnoticed [4].

Data Preparation: In the preparation stage, AI adds value by reducing much of the tedious work required to make uncleaned raw data usable, producing consistent data faster. Google’s Knowledge Catalog can automate data enrichment by mining schemas, logs, BI models, and unstructured signals, and its auto data quality features can generate or reuse rules, execute scans, and integrate results with monitoring and alerting [6]. The prepared data is more

dependable, which is important when the same datasets will feed dashboards, models, and AI agents.

Data Classification: AI can be particularly adept at recognizing patterns. This is useful at the classification stage, particularly when dealing with substantial amounts of data. Microsoft Purview's machine learning capabilities can classify content and automatically apply appropriate policies [5], while AWS Macie uses machine learning, pattern matching, managed data identifiers, custom data identifiers, and allow lists to detect sensitive data [4]. Organizations can label and route information faster and more consistently than with manual reviews [4], [5].

Storage, Monitoring, and Governance

Data Storage: AI improves storage by making stored data easier to understand, search, and retrieve. AI-enabled tools can scan stored assets, identify sensitive content, enrich metadata, and flag exposure risks [4].

Data Quality Monitoring: AI can provide significant value here by continuously monitoring data quality as opposed to relying on manual checks. Google's auto data quality capability allows teams to define and measure table quality, automate scans, validate data against built-in or custom rules, generate rules from profiling, and integrate failures with logging and alerting [6]. Knowledge Catalog also allows for policy-based quality checks and anomaly detection across data sources [6]. Quality problems are typically cheaper to solve when detected early, rather than further downstream once poor-quality data becomes evident and negatively impacts processes and decisions. AI-assisted quality checks make freshness, completeness, and rule compliance much more evident [6].

Data Governance: AI is strategically important for data governance, as it can connect metadata, lineage, quality, and policy. NIST's AI RMF is built around GOVERN, MAP, MEASURE, and MANAGE, with governance intentionally designed as the function that informs the others [2]. AI can surface ownership context, connect upstream and downstream dependencies, and help enforce consistent rules [2], [6]. That means fewer governance decisions made with little information and more decisions made with evidence, traceability, and policy context [2].

Retention, Archiving, and Disposal

Data Retention: AI is useful in data retention because retention rules are often broad, repetitive, and difficult to apply consistently across large volumes of data. AI can help classify records, identify content subject to retention policies, flag items nearing review, and reduce unnecessary long-term storage [5]. Microsoft Purview supports classification, retention, disposition review, and lifecycle policy management [5]. The benefit is both operational and compliance-related, as organizations can retain required information more consistently while

reducing the risk of keeping outdated, redundant, or sensitive data longer than necessary [5]. However, AI should only support retention decisions rather than fully replace human oversight where legal holds, regulatory requirements, or business-critical records are involved.

Data Archiving: AI improves archiving by helping determine which data still has business, legal, or analytical value and by preserving useful metadata [7]. This makes older information easier to retrieve, interpret, and connect to its original business context [7]. Instead of becoming “cold but forgotten” data, archived records remain searchable and better organized through classification, lineage, and metadata enrichment [7]. The benefit is stronger long-term accessibility and reduced confusion when archived data must later be reviewed, reused, or audited [7].

Data Disposal: AI supports disposal by identifying records that appear eligible for deletion, assembling evidence, and helping reduce over-retention [5]. This is valuable because defensible deletion requires more than simply removing files, as organizations must know what was deleted, why it was eligible, and whether any legal or business reason required it to be kept [5]. However, final disposal decisions should remain governed by human-approved policies because wrongful deletion can create legal, regulatory, or audit risk [2].

3.2 Drawbacks

Collection, Preparation, and Classification

Data Collection: This is where many downstream problems originate [8]. Potential concerns include bias in source selection, feedback loops, data poisoning, collection of irrelevant or low-quality data, privacy violations during ingestion, and over-representation of easily accessible sources [8]. As Sambasivan et al. note, the entire concept of “data cascades” means that problems introduced during data collection often remain hidden until much later in deployment, where they may manifest as performance issues, failures, or harms [8]. The authors of the paper repeatedly discuss practitioners struggling with collecting representative data and relying on whatever data was available [8]. Everything from using data from one geographic region and attempting to generalize elsewhere without prior consideration, the reuse of data that was not collected for AI purposes, working with incomplete datasets because no alternatives existed, and using non-representative sampling because of practical constraints all have the potential to lead to issues further along in the data lifecycle [8]. In addition to the data issues, the data collection infrastructure itself has the possibility of being unreliable [8]. The paper describes camera movement, sensor drift, environmental changes, behavioral changes, and regulatory changes as factors that can produce these cascades [8]. If the collection process were delegated solely to AI oversight, such degradation would likely

propagate much further than it would under a human-led intake process before the issue was caught [8]. In short, AI can scale collection processes far beyond human capabilities, but it can also scale collection mistakes far beyond human oversight [8].

Data Preparation: This stage carries subtle but significant risks [8]. AI may remove information that appears anomalous but is actually important, normalize away edge cases, impute incorrect values, deduplicate records that should remain distinct, and introduce errors through automated transformations [8]. Both Sambasivan et al. and the International Organization for Standardization note that data is routinely modified based on assumptions [3], [8]. The paper in particular describes situations where teams corrected values, merged datasets, discarded records, or restarted collection efforts without sufficient domain expertise [8]. The ISO standards help to mitigate this, so the problem is less prevalent than it once was [3]. But as noted in the Data Collection section, having AI as part of the process greatly increases the odds that even minute deviations, misunderstandings, omissions, or errors could have significantly higher impact before any issue is detected [8]. This occurs either because the AI proceeds at speeds and scales that are difficult for humans to process, or because the AI itself missed or assumed something which was not caught or communicated (a habit for which LLMs are notorious). One recurring theme from Sambasivan et al. is that datasets often lacked metadata timestamps, collection conditions, geographic context, or collection methodology [8]. When metadata was missing, the data engineers and computer scientists frequently assumed there was an issue with the data, or discarded it altogether [8]. One of the paper's strongest findings was that technical teams often lacked the application-domain expertise needed to properly interpret data [8]. The authors describe inaccurate ground truth, subjective labels, missing context, and poor representation as a result of this [8]. The paper further details that these issues were often only discovered later through manual review or deployment failures [8]. The more aggressively AI "improves" data, the greater the risk that it alters the original meaning of the dataset [8].

Data Classification: This is the area where most people would likely default to an automated system. Potential issues include hallucinated metadata, misclassification, label drift, bias, and inconsistent application of taxonomy or features [3], [9]. For example, an LLM generating tags for documents may create plausible, but false, classifications that then become part of the organization's official metadata [9]. ISO/IEC 25012 states that data should correctly represent the real-world entity or concept it describes, should be represented uniformly across datasets and systems, should be regarded as trustworthy and believable, and its lineage and origins should be understandable [3]. Ribeiro et al. argue that if an AI classifier generates incorrect tags, labels, or metadata, the resulting metadata is no longer accurate [9]. This can directly lead to hallucinated metadata and/or misclassification [9]. An LLM, for example, may generate metadata that appears plausible but does not accurately describe the underlying document [9].

If AI systems apply taxonomies inconsistently, the same document could be categorized differently on different runs, different labels might get assigned to similar records, and classification rules may change after retraining [9]. When assessing credibility, if users discover repeated classification errors, confidence in the metadata will deteriorate [3], [9]. Ribeiro et al. note this may introduce unintended bias on the part of the operators, because even plausible-looking classifications become less credible if their generation process cannot be validated [9]. This has the potential to create a situation where users may not understand AI-generated metadata because they cannot discern why a label was assigned, what features were used, or how a classification was produced [9]. While ISO/IEC 25012 does not specifically address AI, this is a situation that it cautions against with respect to data consistency and understandability [3]. This remains a uniquely AI-era problem.

Storage, Monitoring, and Governance

Data Storage: This stage is frequently overlooked. Potential issues include opaque retrieval prioritization, explainability problems, hidden biases in indexing, and difficulty auditing AI-generated storage structures [2], [9]. The NIST AI Risk Management Framework (AI RMF 1.0) emphasizes that explainability is a core element of responsible AI [2]. One of the challenges with giving an AI a decision-making role in data storage is that machines do not optimize the same way humans do. A situation could arise where the system organizes data storage according to its ‘preferences’ or training instead of how a person might logically organize it. The AI’s rationale would likely emphasize retrieval speed and resource minimization, optimizations that do not necessarily make the data easy for a person to find or use. Ribeiro et al. argue that AI systems often operate as black boxes whose decision-making processes are difficult to understand [9]. While AI-assisted storage and retrieval systems may improve efficiency, opaque indexing, prioritization, and/or linking decisions can make it difficult to determine why information was organized the way it was [9]. The authors further note that trust in AI systems requires explainability, as users must be able to understand and evaluate the reasoning behind automated decisions before those decisions can be reliably audited or trusted [9]. The fundamental question that comes out of this is: if an AI decides where data should reside, can a human explain why a specific record was stored, linked, or prioritized [2], [9]? If that question cannot be answered satisfactorily, there is a high probability the system will not be trusted [9].

Data Quality Monitoring: AI is often promoted as the solution for this stage [2]. But false alarms, missed anomalies, model drift, alert fatigue, and dependence on training data quality often sabotage the system in unanticipated ways [2], [8]. Sambasivan et al. highlight that data problems often remain invisible until deployment [8]. As noted previously, the authors repeatedly emphasize that data cascades are opaque, delayed, difficult to detect, and only visible through performance degradation [8]. The NIST AI RMF reinforces this with particular emphasis on data quality management, reiterating that this is a cycle during the lifetime of a

project, not something that is assessed once and then forgotten [2]. One of the paper's key criticisms, and an implicit caution from the NIST AI RMF, is that AI teams tend to focus heavily on model metrics while having few mechanisms to evaluate the underlying quality of the data itself [2], [8]. Sambasivan et al. document hardware drift, environmental drift, and human drift; due to this, the authors argue good metrics do not necessarily indicate good data [8]. Put another way, high model performance does not guarantee high-quality data, making it difficult for monitoring systems to detect underlying data quality degradation [8]. These changes alter incoming data and degrade system performance over time [8]. If AI is a participant in the data management lifecycle, it may well fall victim to the same issues that plague the system it is supposed to be monitoring and supporting, since the system responsible for monitoring data quality may itself experience quality degradation over time [2], [8].

Data Governance: This is where Human-Centred AI and AI Governance become relevant [2]. Potential concerns like lack of transparency, accountability gaps, regulatory compliance uncertainty, and difficulty auditing decisions present substantial challenges, as outlined elsewhere in this paper [2], [9]. The NIST AI RMF repeatedly states that accountability cannot be delegated to AI [2]. One of the central themes of the AI RMF is that governance structures must establish clear accountability for AI systems and their outcomes [2]. It emphasizes that governance functions, assigned responsibilities, oversight mechanisms, and organizational accountability are core to the framework [2]. AI risk is treated as an organizational responsibility rather than a technical problem alone [2]. The AI RMF further identifies transparency, interpretability, and explainability as important characteristics of trustworthy AI systems [2]. NIST argues that stakeholders should be able to understand system capabilities, limitations, intended uses, and decision processes to a reasonable degree [2]. The RMF repeatedly emphasizes continuous monitoring and risk management throughout the AI lifecycle, and consistently advocates for human oversight mechanisms appropriate to the context and level of risk [2]. Organizations must continually evaluate model performance, compliance, emerging risks, and unintended consequences, particularly when AI systems are making governance-related recommendations [2]. The recurring question of who is responsible when an AI-driven governance system makes an incorrect decision remains as relevant here as with any other AI application [2].

Retention, Archiving, and Disposal

Data Retention: This stage introduces additional legal risk [2]. Potential concerns include retaining data longer than necessary, premature deletion, failure to recognize legal hold requirements, and misinterpretation of business value [2], [3]. Both the NIST AI RMF and ISO/IEC 25012 recommend that data conform to applicable regulations, policies, and standards [2], [3]. Retention decisions often have legal requirements, regulatory requirements, and contractual requirements, which are not unique to AI. But if AI is participating in the retention

process, there is a risk that it makes an incorrect assessment about a record. If we examine a hypothetical situation in which an AI system determines a record appears unnecessary and recommends deletion, or worse, has authority for automatic deletion, we run into a situation where compliance requirements may demand that the data be retained regardless of apparent utility [2], [3]. In that case, the AI would have caused a regulatory or compliance breach. Further, both ISO and NIST recommend data be sufficiently up-to-date for its intended use [2], [3]. If an AI retention system employs current usage patterns as a metric to evaluate the value of a given set of data, historical data may no longer appear "current" [3]. This could cause the system to deprioritize or even remove such data [3]. This may have legal impact, unintentionally impede auditability, or impact the research value of the dataset [2], [3]. Alongside this is the notion that retention decisions should be explainable [2], [9]. Both frameworks suggest that if data requires deletion or retention, those responsible for the data should be able to answer 'Why?', 'Based on which criteria?' and 'Can the recommendation be audited?' [2], [3]. Failing to do this could raise concerns around legal holds and regulatory review [2]. These problems would be even worse if an AI were making the decisions, because without auditability of the model itself, those answers may be buried behind several layers of black boxes [9]. AI can estimate usefulness, but AI cannot necessarily understand future legal significance [2].

Data Archiving: This is subtle but important [6]. Underestimating future value, historical bias, and loss of rare edge-case information are just some of the possible issues here [7]. Both the ISO/IEC 25012 and NIST frameworks state a preference that archived information should remain retrievable when required [2], [3]. The catch with an AI archiving system is that it may prioritize commonly accessed data while undervaluing rare but important records, making finding or accessing those records more challenging [2]. ISO emphasizes maintaining data quality throughout the lifecycle [3]. Aggressive archiving strategies governed by predictive models may remove context, remove metadata, or discard supporting records [3], [7]. This could cause archived data to remain present but become completely useless [3]. Additionally, future users need to understand where data came from, why it was archived, and how it relates to other records [3], [7]. If AI-driven archiving makes these decisions automatically without sufficient documentation, the user searching for the data faces a significant structural obstacle [7]. One of the reasons to archive data is because historical records often become more valuable over time [7]. Similar to the Data Retention problem, an AI system trained on current usage patterns may conclude that nobody accesses the data while overlooking its future value for investigations, audits, historical analysis, retraining models, or legal proceedings [2], [7]. This is essentially another credibility and preservation problem and impacts users in the same way [3], [7]. A model trained on current usage patterns may conclude the data does not get accessed, but that does not mean the data has no future value [7]. This kind of future value is an abstract criterion that AI systems struggle to gauge [7].

Data Disposal: This is potentially the highest-consequence stage [2]. Irreversible deletion, regulatory violations, incomplete deletion, and failure to verify downstream dependencies are all almost impossible to undo [2]. The NIST AI RMF notes that assessing both the likelihood and severity of harm is paramount when evaluating AI risks [2]. When applied to disposal, a classification error may be correctable later, but a deletion error has a high probability of not being recoverable [2]. The challenge with AI participation in this stage is similar to the other areas of the data lifecycle, with the added complication that traceability may be even more difficult to ascertain, as the element that the AI system has applied its logic to may no longer exist [2]. In most other cases, one could go back to look at the raw data to see what path the AI took in its reasoning [9]. If the data no longer exists, there is a high probability that the AI likewise retains no trace of that data element either, rendering any attempt at an audit impossible [2], [9]. Although NIST does not discuss data deletion specifically, the framework repeatedly emphasizes applying greater oversight to higher-risk AI applications [2]. Data disposal fits naturally into this category because deletion is often irreversible, regulatory requirements apply, and business impacts may emerge years later, potentially long after the system that made the decisions has been retired [2]. NIST's emphasis on lifecycle risk management implies that governance responsibilities extend through the end-of-life phase of information management, including disposal and deletion activities [2]. For this reason, this area of the data lifecycle is especially sensitive to mismanagement by an AI participant [2]. Unlike a classification error, deleted data may simply be gone, and retention violations cannot be undone just because the threshold was missed [2].

4. Findings: Governance of AI in the Data Management Lifecycle

Data governance should work as a framework that runs through every phase of the data lifecycle, rather than a checkpoint applied after the fact. Because automated tools operate at speeds that outpace manual oversight, governance sets the ground rules for how those tools, the organization's systems, and human supervisors interact [2].

Collection, Preparation, and Classification

Data Collection: Preventing data cascades starts with clear intake rules that limit what automated collection tools are allowed to gather [8]. In line with the NIST AI RMF, organizations should document where their data comes from instead of letting automated systems pull data from unverified sources [2], [8]. Governance rules should also require metadata to be captured at the moment of ingestion, so that timestamps, geographic context, and the original collection conditions stay attached to each record [8]. This guards against what Sambasivan et al. call "data cascades," where hidden errors at intake lead to delayed and costly failures later on [8].

Tools like Amazon Macie can be placed inside data pipelines so that human reviewers can audit automated ingestion before bad data spreads downstream [4].

Data Preparation: Automated data enrichment and cleaning should not happen inside an unmonitored black box [9]. At the preparation stage, any algorithmic changes, such as filling in missing values, deduplication, or removing outliers, should follow reproducible, rule-based settings [9]. ISO/IEC 25012 states that data must accurately represent real-world concepts, a requirement that is easily broken when automated tools discard or alter anomalies without domain expertise [3], [8]. Governance should therefore require all system transformations to be logged, so that data engineers can reconstruct the original dataset if an automated model changes its meaning [3].

Data Classification: Automated classifiers need their taxonomies validated before they are trusted [9]. When platforms like Microsoft Purview or Google Cloud Knowledge Catalog apply classification tags automatically, governance policies should set clear confidence thresholds [5], [6]. Any classification that falls below the threshold should be routed to a human for manual review [9]. This human-in-the-loop check prevents the spread of "hallucinated metadata" and limits label drift, protecting the credibility and auditability standards described in ISO/IEC 25012 [3], [9].

Storage, Monitoring, and Governance

Data Storage: Governance in AI-driven storage is about keeping machine-optimized indexing understandable to the people who manage it [9]. Because storage systems often organize data around mathematical efficiency rather than how humans would group it, governance should enforce explainability standards [9]. Under the NIST AI RMF, stakeholders should be able to audit and explain why a particular record was prioritized, linked, or stored where it was [2], [9]. Indexing systems should follow open metadata standards so that compliance officers can trace data locations without hitting opaque algorithmic barriers [2], [9].

Data Quality Monitoring: Monitoring systems need protection against their own drift [8]. Because monitoring algorithms are sensitive to hardware updates, environmental shifts, and changing data profiles, governance should schedule periodic validation of the monitoring tools themselves [8]. The NIST AI RMF treats risk assessment as a continuous loop rather than a one-time milestone, so organizations should cross-check automated quality metrics against manual baselines [2]. This step ensures the monitoring tool does not degrade silently, which would otherwise hide deeper data corruption [2], [8].

Data Governance: At the top level, governance makes clear that accountability stays with people and cannot be handed off to an algorithm [2]. Tools like Google Cloud Knowledge Catalog can map data lineage, connect related datasets, and flag potential compliance

violations, but they remain decision-support tools [6]. Under the NIST AI RMF, final authority and legal liability for data policies rest with designated human owners [2]. Governance bodies are responsible for ongoing oversight of the lifecycle tools themselves, auditing their performance, fairness, and transparency [2].

Retention, Archiving, and Disposal

Data Retention: Automated retention processes must follow legal, regulatory, and contractual requirements [2], [3]. Because machine learning tools judge data value by how often it is accessed, they often miss the long-term legal significance of dormant records [2]. Governance policies should prevent AI from carrying out automatic deletions during retention reviews [2]. Instead, systems like Microsoft Purview should be set up to generate disposition recommendations, with final deletion sign-off left to compliance officers [5]. This human check protects data under legal holds and aligns with the retention requirements of ISO/IEC 25012 [2], [3].

Data Archiving: Archival governance is about preserving context along with the data [6]. When algorithms decide which records move to cold storage, governance rules should make sure that lineage, original metadata, and operational context are preserved alongside the data itself [7]. Following both ISO/IEC 25012 and recent archival research, archived data should remain discoverable and understandable for future users [3], [7]. Governance controls should also stop automated systems from over-aggressive compression or truncation, which can permanently strip away rare edge cases needed for future audits or model retraining [3], [7].

Data Disposal: Because deletion is irreversible, disposal requires the strictest oversight of any stage [2]. Automated deletion agents are a serious compliance risk because once a record is erased, there is nothing left to trace [2]. To reduce this risk, governance frameworks should require tamper-proof deletion logs [2]. These logs should record what data was removed, the policy justification for removing it, and who approved it [2]. In line with the NIST AI RMF's principle of matching oversight to risk, final disposal steps should require a human sign-off before data is permanently wiped [2].

4.1 Recommendations

Based on the findings of this research, several recommendations emerge for organizations seeking to implement AI within data lifecycle management processes:

- **Establish Human-in-the-Loop Governance:** AI systems should function as decision-support tools rather than autonomous decision-makers, particularly for retention, compliance, and disposal activities.

- **Adopt Formal Governance Frameworks:** Organizations should align AI-enabled data management practices with recognized standards such as the NIST AI Risk Management Framework and ISO/IEC 25012 to improve accountability, transparency, and compliance.
- **Implement Continuous Monitoring and Validation:** AI systems used for classification, quality monitoring, and governance should undergo periodic reviews to detect model drift, performance degradation, and emerging risks.
- **Maintain Comprehensive Metadata and Lineage Tracking:** Strong metadata management practices improve traceability, auditability, and trustworthiness throughout the lifecycle.
- **Restrict Fully Automated Data Disposal:** Given the irreversible nature of deletion activities, final approval authority should remain with designated human stakeholders.

5. Conclusion

The integration of artificial intelligence into data lifecycle management provides clear operational efficiency, transforming manual data routines into continuous, automated pipelines [5], [6]. As the findings on benefits in Section 3 highlight, automated systems dramatically accelerate metadata ingestion, improve classification accuracy, and enable real-time quality monitoring [5], [6].

However, as the findings on drawbacks in Section 3 demonstrate, these tools introduce complex technical vulnerabilities, including opaque storage indexes, model drift, and systemic data cascades that originate silently during collection [8], [9]. These risks were identified by comparing academic case studies against formal standards and frameworks, which made it possible to trace how failures introduced at one lifecycle stage propagate into later stages.

Ultimately, this report shows that AI tools cannot operate as fully autonomous actors [2], [8]. Data safety cannot be achieved by relying only on a model's internal metrics; it requires a human-led governance framework that covers every phase of the pipeline [2], [8]. In compliance with the NIST AI Risk Management Framework and ISO/IEC 25012 standards, organizational accountability must remain entirely human, positioning automated tools as decision-support infrastructure rather than primary decision-makers [2], [3].

Enterprise architectures from AWS, Google, and Microsoft can successfully manage complex data sorting and alerting workloads across all nine lifecycle stages, but human operators must keep final sign-off over high-risk decisions such as legal retention, model validation, and irreversible data disposal [2], [4], [5], [6], [7].

Therefore, the role of AI in data lifecycle management is fundamentally a tradeoff between scalability and control, requiring governance mechanisms to compensate for the amplification of lifecycle risks.

6. References

- [1] Southern Alberta Institute of Technology, "Research Paper: AI Data Management and Quality Assurance," Course Assignment Guidelines (409), SAIT, Calgary, AB, Canada, 2026.
- [2] National Institute of Standards and Technology, *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*, NIST AI 100-1, U.S. Department of Commerce, Gaithersburg, MD, USA, Jan. 2023. Available: <https://doi.org/10.6028/NIST.AI.100-1>.
- [3] *Software Engineering—Software Product Quality Requirements and Evaluation (SQuaRE)—Data Quality Model*, ISO/IEC Standard 25012:2008, 2008.
- [4] Amazon Web Services, *Amazon Macie, Discovering sensitive data with Macie, Monitoring data security and privacy with Macie, and Use Macie to discover sensitive data as part of automated data pipelines*, AWS Documentation, Seattle, WA, USA, 2025.
- [5] Microsoft, *Microsoft Purview and Microsoft Purview Data Lifecycle Management*, Microsoft Security Documentation, Redmond, WA, USA, 2025.
- [6] Google Cloud, *Knowledge Catalog, Auto data quality overview, and About data lineage*, Google Cloud Documentation, Mountain View, CA, USA, 2025.
- [7] G. Shinde, T. Kirstein, S. Ghosh, and P. C. Franks, "AI in Archival Science: A Systematic Review," *arXiv preprint arXiv:2410.09086*, Oct. 2024.
- [8] N. Sambasivan, S. Kapania, H. Highfill, D. Akrong, P. Paritosh, and L. M. Aroyo, "'Everyone wants to do the model work, not the data work': Data cascades in high-stakes AI," in *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, Yokohama, Japan, May 2021, pp. 1–15. Available: <https://doi.org/10.1145/3411764.3445518>.
- [9] M. T. Ribeiro, S. Singh, and C. Guestrin, "'Why should I trust you?': Explaining the predictions of any classifier," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, San Francisco, CA, USA, Aug. 2016, pp. 1135–1144. Available: <https://doi.org/10.1145/2939672.2939778>.

7. AI Usage Statement

Artificial intelligence tools were utilized during the preparation of this paper to support literature structuring, stylistic harmonization, alignment with standard technical guidelines, and citation management across the project lifecycle. All content, findings, and structural analyses were fully reviewed, verified, and approved by the collective authors to ensure academic and professional accuracy.